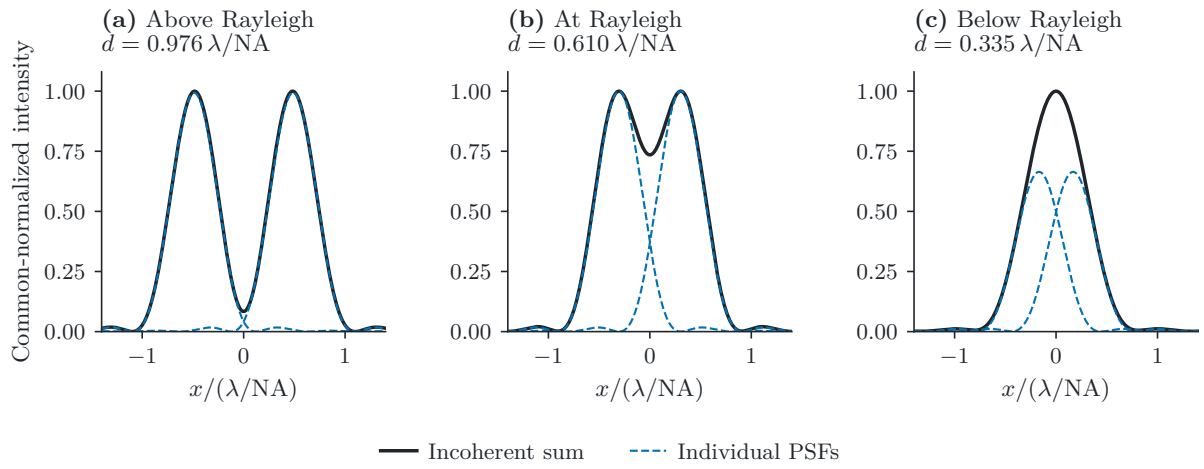


Theory notes – Figure 1

Two-point Rayleigh criterion



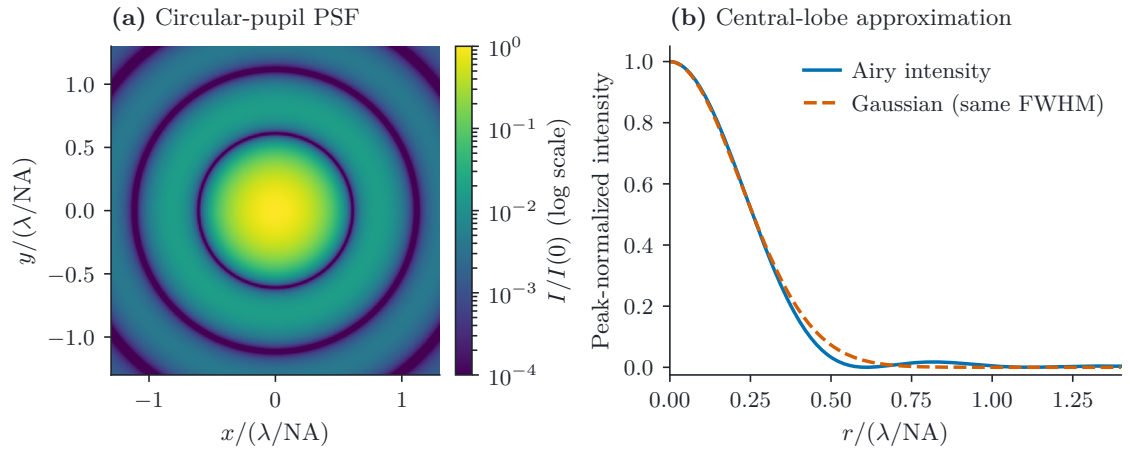
Caption. Two equal-brightness incoherent point sources through an ideal circular pupil. Each panel uses one normalization for the sum and both component PSFs, so the components add exactly. Rayleigh separation is $0.610 \lambda/\text{NA}$; it is a conventional two-point criterion, not a universal localization limit.

Document locations. Theory notes, Figure 1; Manuscript, Figure 2

Evidence type. analytical or synthetic teaching example

Theory notes – Figure 2

Airy PSF and Gaussian approximation



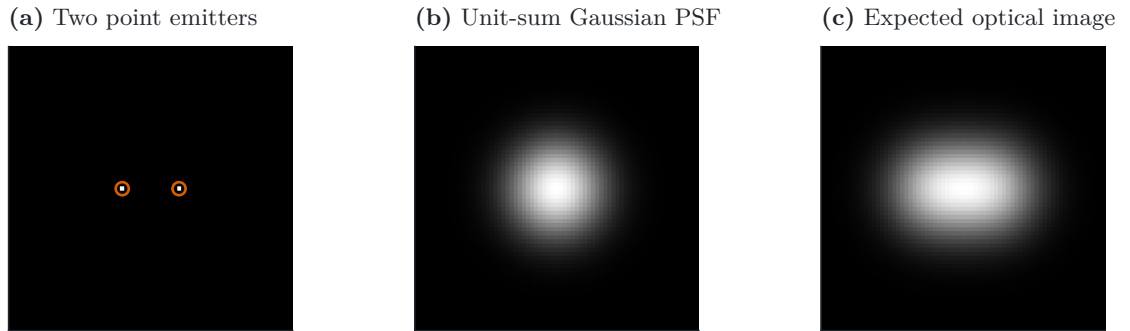
Caption. Ideal scalar, incoherent circular-pupil PSF. The image uses a labelled logarithmic colour scale to reveal rings; the profile is linear. The Gaussian matches the Airy FWHM only, not the side lobes or total energy.

Document locations. Theory notes, Figure 2; Manuscript, Figure 1

Evidence type. analytical or synthetic teaching example

Theory notes – Figure 3

Incoherent image formation



$g = f * h$; camera pixels integrate g over their area. Images scaled separately for visibility.

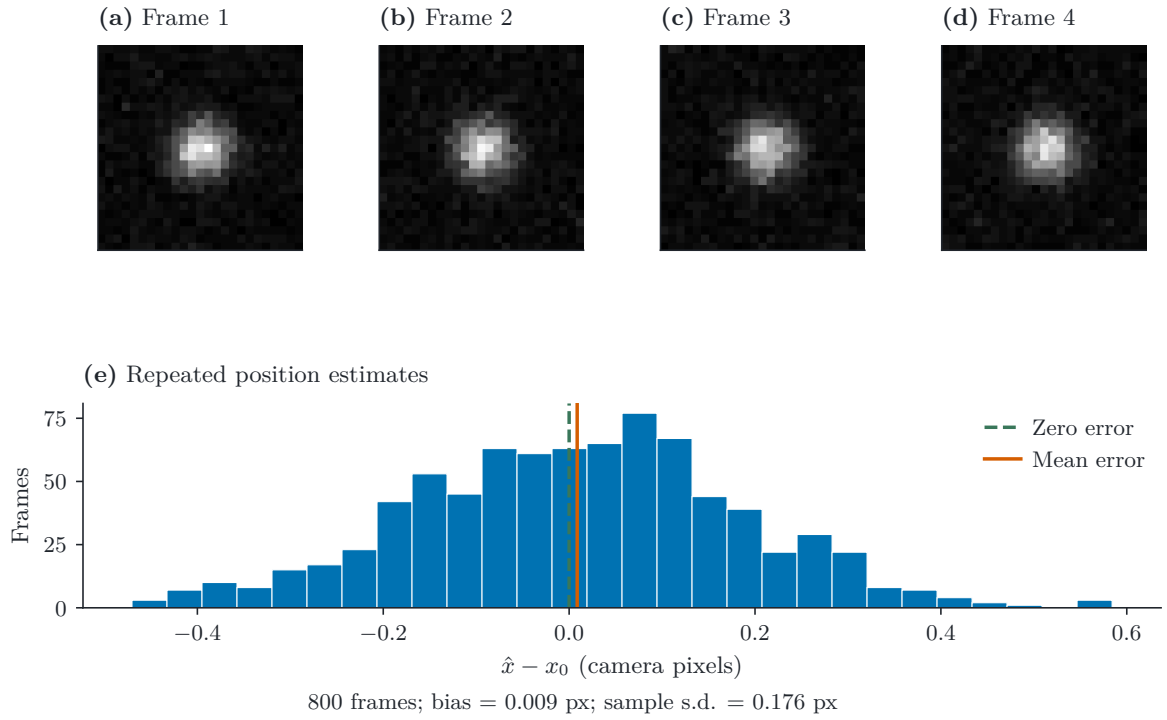
Caption. Linear shift-invariant, incoherent image formation: two equal point sources convolved with a normalized Gaussian PSF. Pixel integration is a subsequent detector operation. Panel brightness is independently scaled; the circles identify source positions.

Document locations. Theory notes, Figure 3; Manuscript, Figure 3

Evidence type. analytical or synthetic teaching example

Theory notes – Figure 4

Noise and repeated position estimates



Caption. Pixel-integrated Gaussian signal with 2000 expected photons, $\sigma = 2.3$ pixels, and Poisson background mean 3 counts/pixel. The V1.2 single-emitter centroid uses $q = 0.98$ and frame-median subtraction. The histogram measures repeated-estimate spread and bias, not per-event uncertainty.

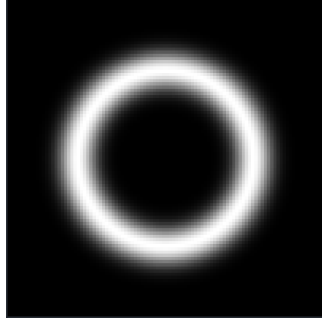
Document locations. Theory notes, Figure 4; Manuscript, Figure 9

Evidence type. analytical or synthetic teaching example

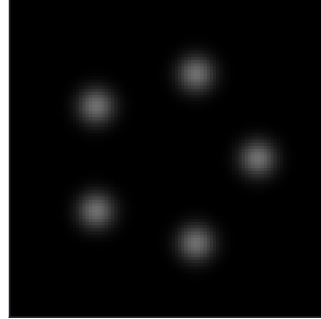
Theory notes – Figure 5

Temporal sparsity

(a) All sources emitting



(b) Five active sources



Same source positions, PSF and intensity scale; only the active subset changes.

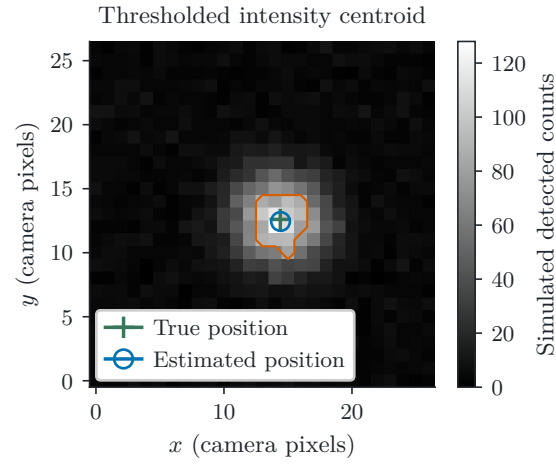
Caption. Temporal sparsity: the same emitter arrangement with all 40 sources active or five well-separated sources active. These are noise-free expectations with one shared intensity scale; sparse activation enables individual localization.

Document locations. Theory notes, Figure 5

Evidence type. analytical or synthetic teaching example

Theory notes – Figure 6

Thresholded intensity centroid



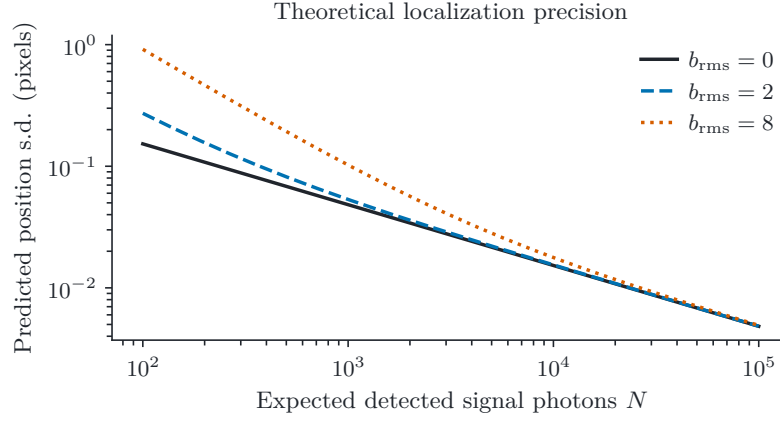
Caption. A synthetic pixel-integrated Gaussian spot with Poisson noise. The orange contour marks the $q = 0.98$ threshold mask; the plus and circle distinguish truth and the application centroid. Subpixel coordinates do not imply zero error. This example uses no background correction.

Document locations. Theory notes, Figure 6

Evidence type. analytical or synthetic teaching example

Theory notes – Figure 7

Theoretical localization precision



$s = 1.5$ px; pixel pitch $a = 1$ px.
 b_{rms} : RMS background noise (photon-equivalent).

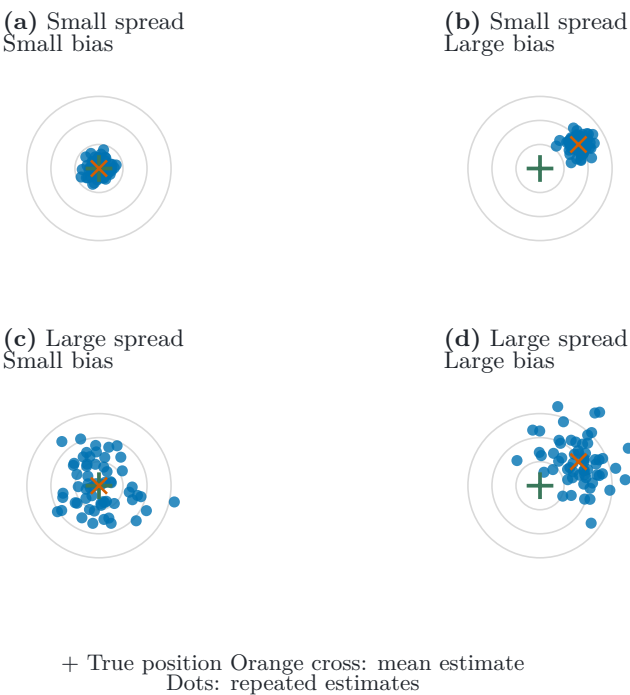
Caption. Thompson-Larson-Webb reference approximation, not uncertainty reported by the centroid software. PSF $\sigma = 1.5$ camera pixels, pixel pitch $a = 1$ in the same length units. b_{rms} is background RMS noise, not mean background. Position s.d. scales as $N^{-1/2}$ in the signal-limited regime and N^{-1} in the background-limited regime.

Document locations. Theory notes, Figure 7; Manuscript, Figure 4

Evidence type. analytical or synthetic teaching example

Theory notes – Figure 8

Spread and bias



Caption. Illustrative samples separate repeatability (spread) from systematic offset (bias). The mean is explicitly controlled. Accuracy depends on both bias and random error; small spread alone is not accuracy.

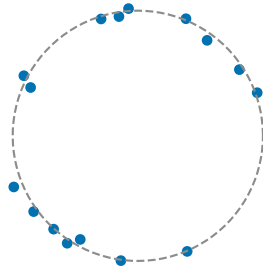
Document locations. Theory notes, Figure 8

Evidence type. analytical or synthetic teaching example

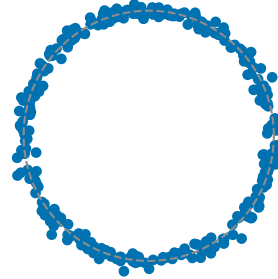
Theory notes – Figure 9

Spatial sampling

(a) Sparse spatial sampling
16 positions



(b) Denser spatial sampling
320 positions



Same positional noise; different spatial coverage.
Repeated detections can revisit the same site.

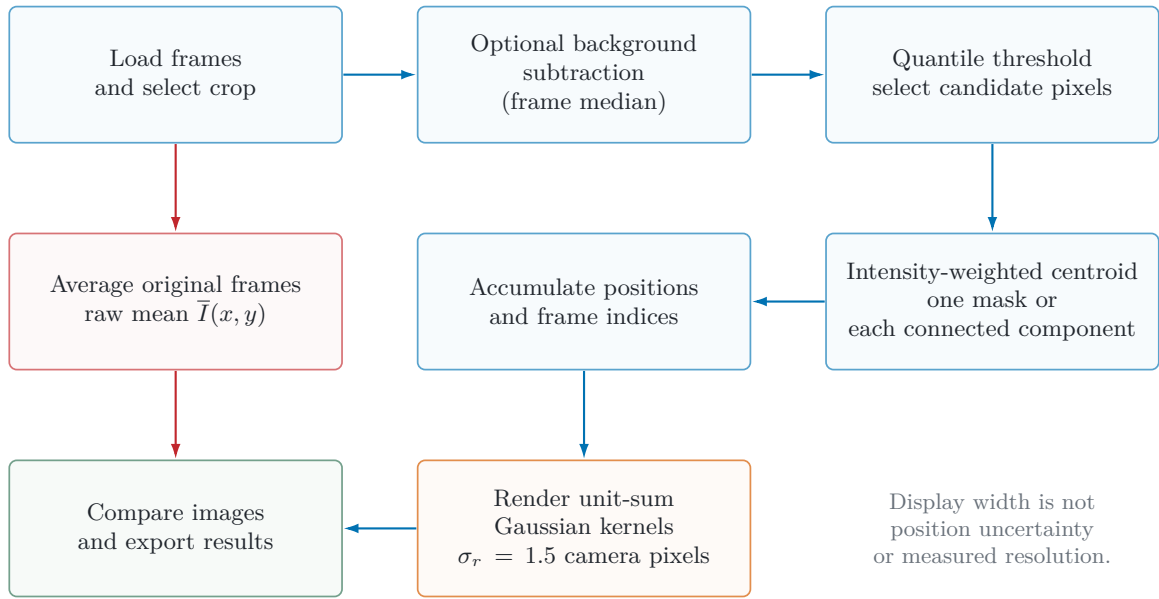
Caption. Synthetic ring sampled at 16 or 320 independently drawn positions with the same radial noise. Sampling spacing and localization precision are different quantities. No universal resolution threshold or adequate sample count is inferred from this illustration.

Document locations. Theory notes, Figure 9

Evidence type. analytical or synthetic teaching example

Theory notes – Figure 10

Reconstruction workflow



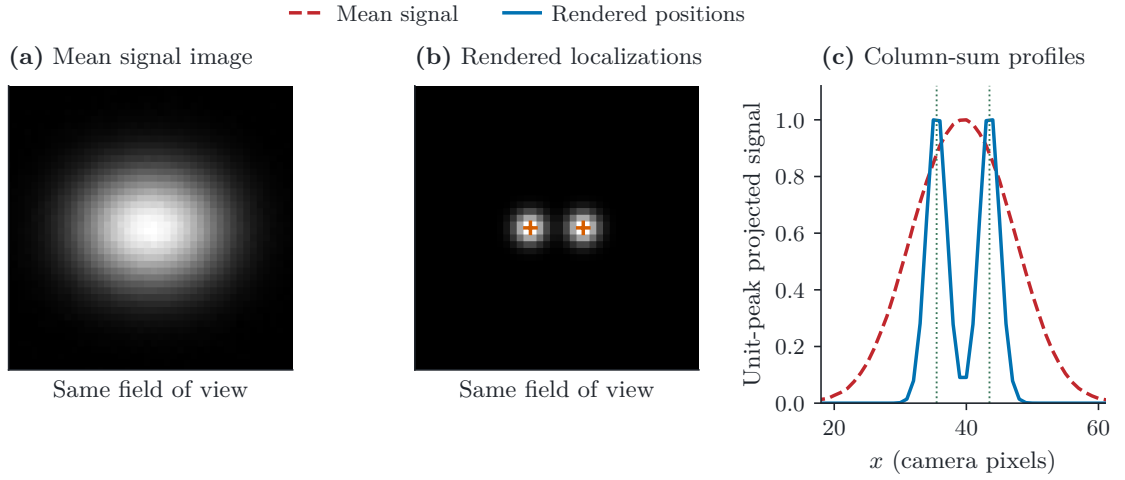
Caption. The current centroid pipeline, with a separate raw-mean branch. Rendering width is fixed and is not an uncertainty estimate.

Document locations. Theory notes, Figure 10; Manuscript, Figure 7

Evidence type. native TikZ explanatory schematic

Theory notes – Figure 11

Synthetic reconstruction



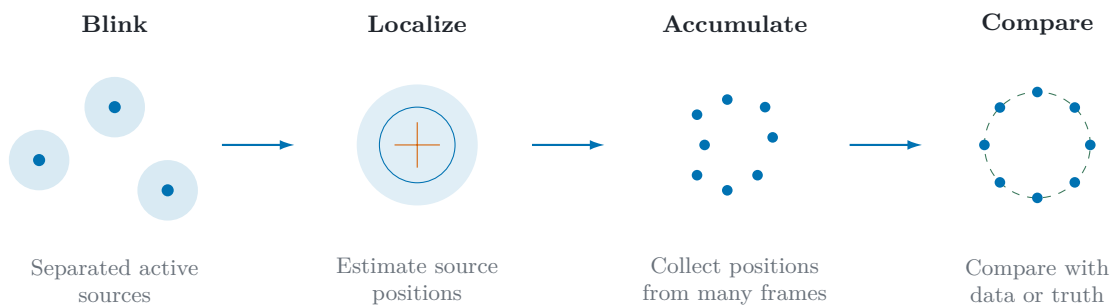
Synthetic sparse blinking; same camera grid. Render width is a display parameter.

Caption. Actual V1.2 centroid reconstruction of 600 synthetic pixel-integrated Poisson frames. Exactly one of two sources is active per frame; separation 8 pixels and optical Gaussian $\sigma = 6$ pixels. $q = 0.995$ with frame-median subtraction. The mean image subtracts the known simulated background for display. Both images and column-sum profiles are independently peak-normalized. Green guides and orange crosses mark truth. Gaussian rendering $\sigma_r = 1.5$ pixels does not measure resolution. Source data and run parameters accompany the figure.

Document locations. Theory notes, Figure 11; Manuscript, Figure 8

Evidence type. synthetic input processed by maintained V1.2 reconstruct_frames

User guide – Figure 1
Localization imaging concept

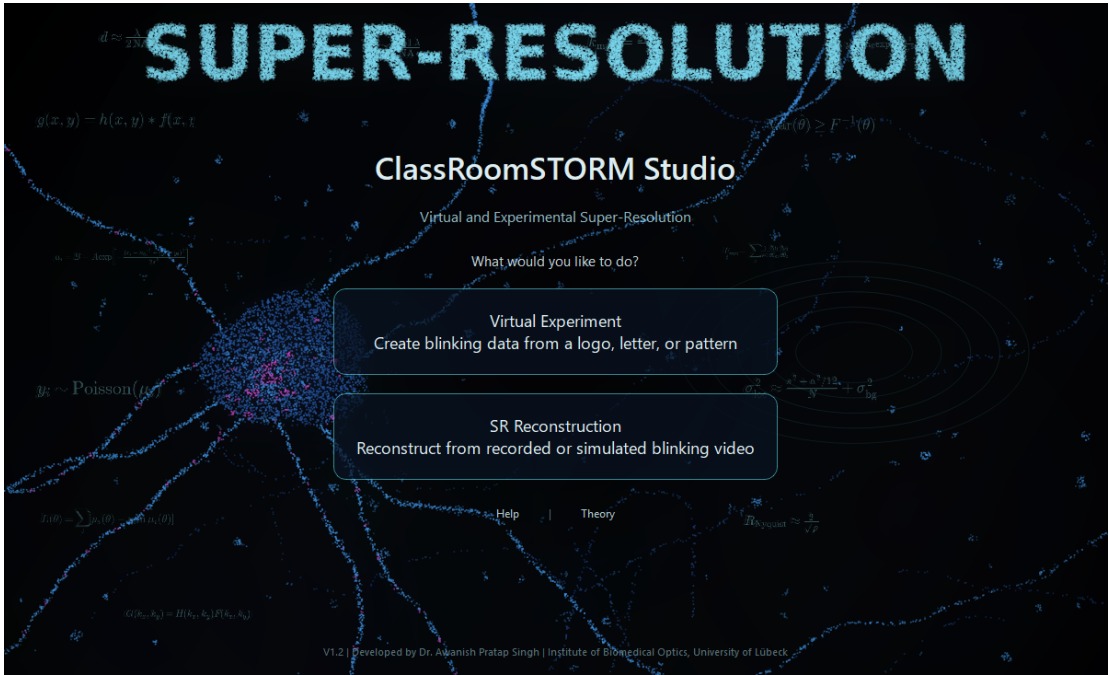


Caption. Conceptual progression from sparse blinking through position estimation to comparison with data or known truth.

Document locations. User guide, Figure 1

Evidence type. native TikZ explanatory schematic

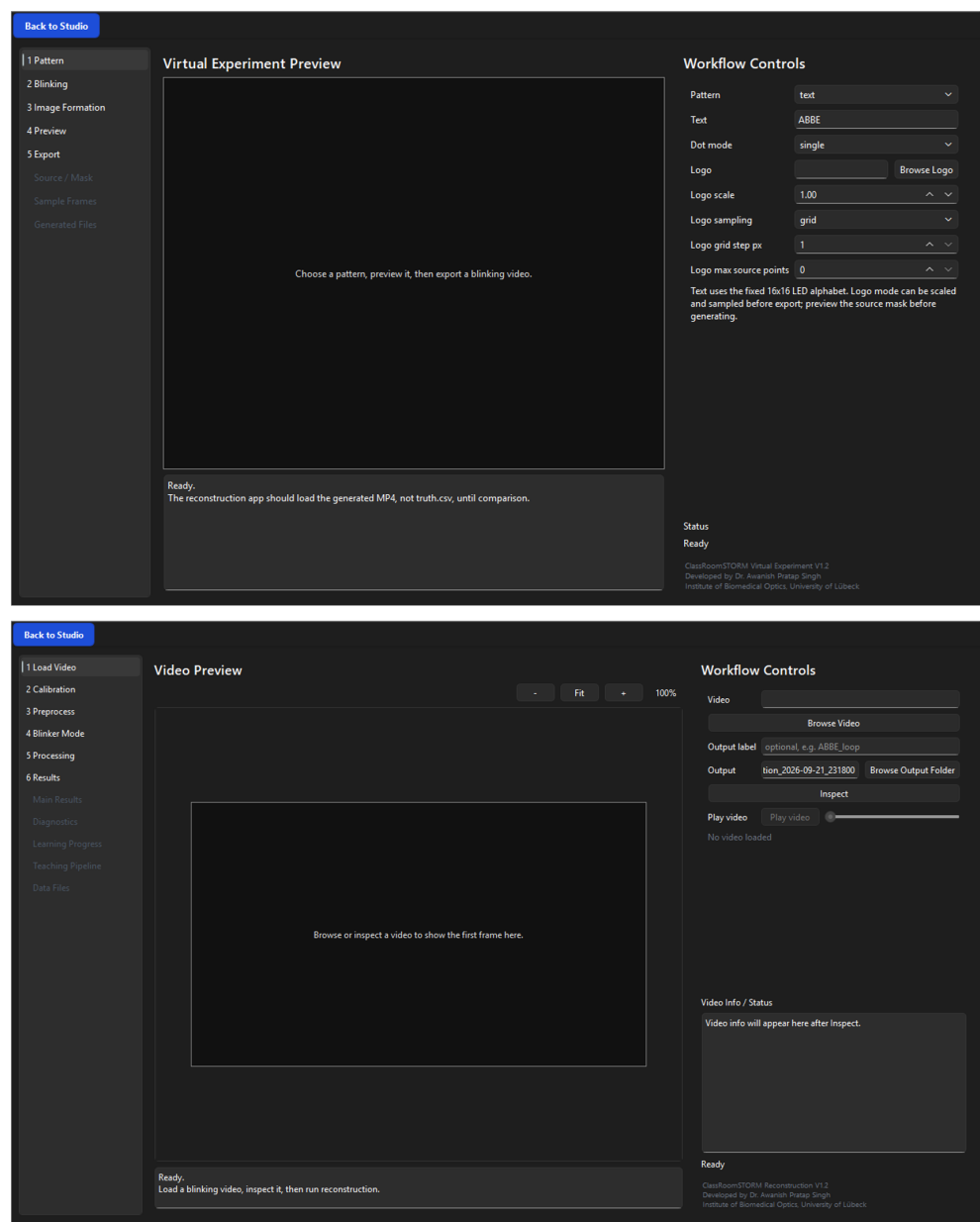
User guide – Figure 2
Studio launcher



Caption. Current Studio launcher, captured from the application.
Document locations. User guide, Figure 2
Evidence type. actual application screenshot

User guide – Figure 3

Quick-start windows



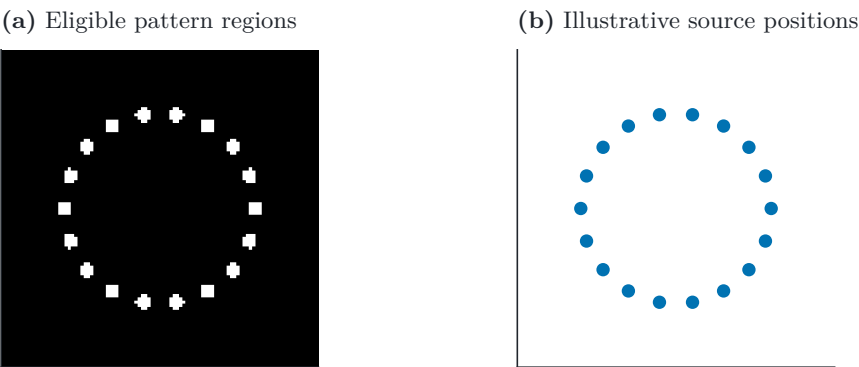
Caption. Current virtual-experiment and reconstruction windows, captured from the application.

Document locations. User guide, Figure 3

Evidence type. actual application screenshot

User guide – Figure 4

Source pattern



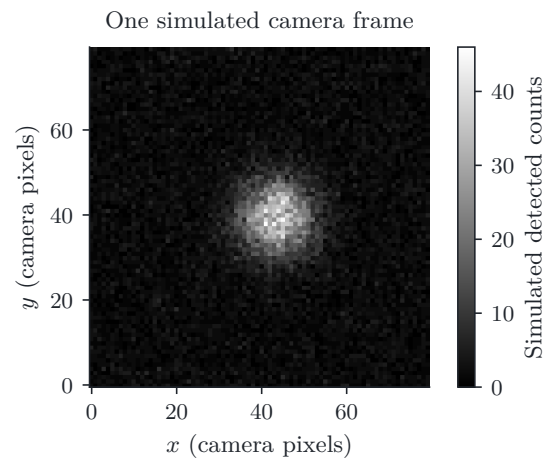
Caption. A schematic source pattern and one illustrative set of source positions. The mask defines eligible regions; it is not a measured optical image or a guarantee of one emitter per bright connected region.

Document locations. User guide, Figure 4

Evidence type. analytical or synthetic teaching example

User guide – Figure 5

Simulated camera frame



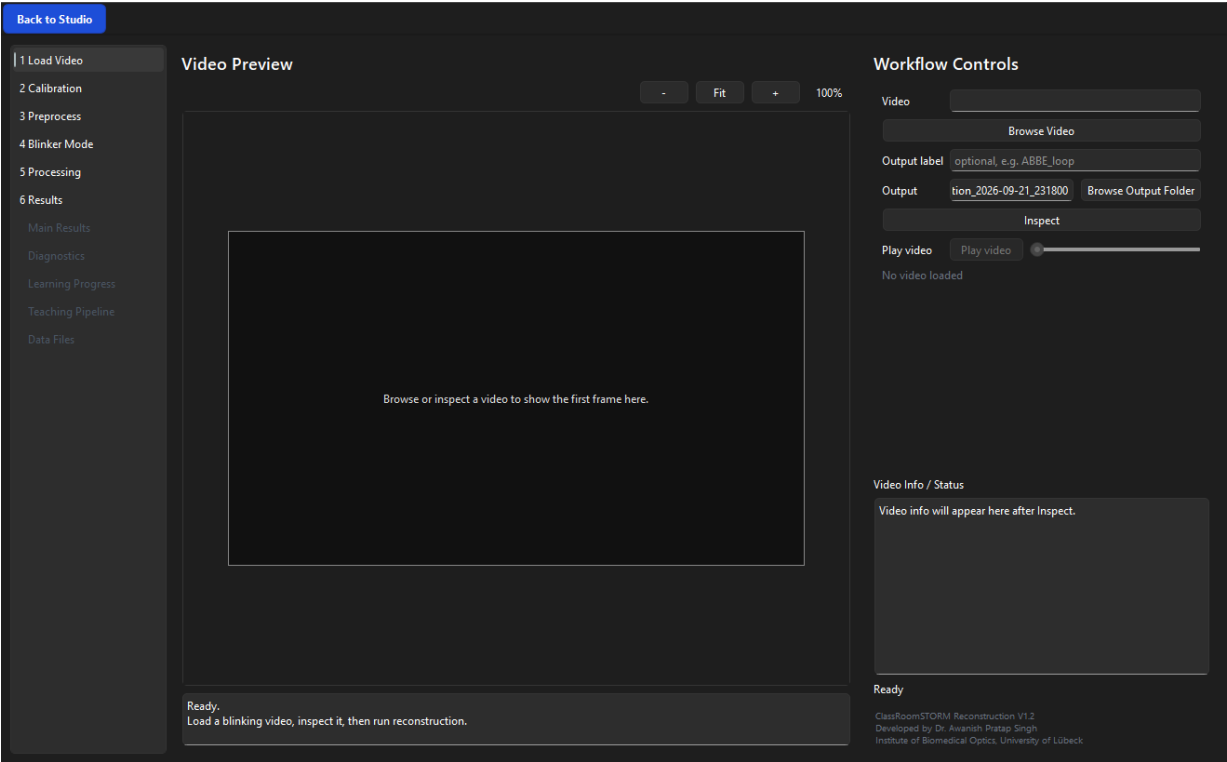
Caption. One raw synthetic frame from the recorded two-source run. Only one emitter is active in this frame; signal and background are Poisson-distributed.

Document locations. User guide, Figure 5

Evidence type. analytical or synthetic teaching example

User guide – Figure 6

Load and crop interface



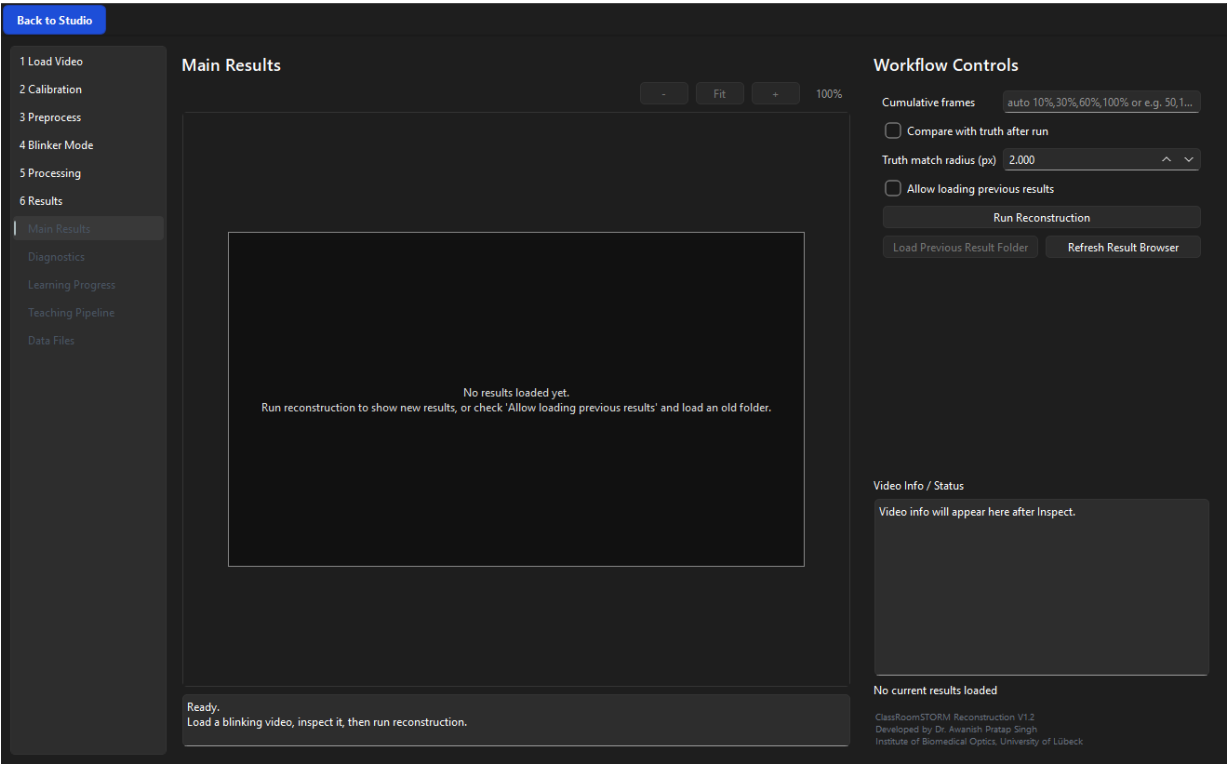
Caption. Current reconstruction load/crop interface before a video is loaded.

Document locations. User guide, Figure 6

Evidence type. actual application screenshot

User guide – Figure 7

Results interface



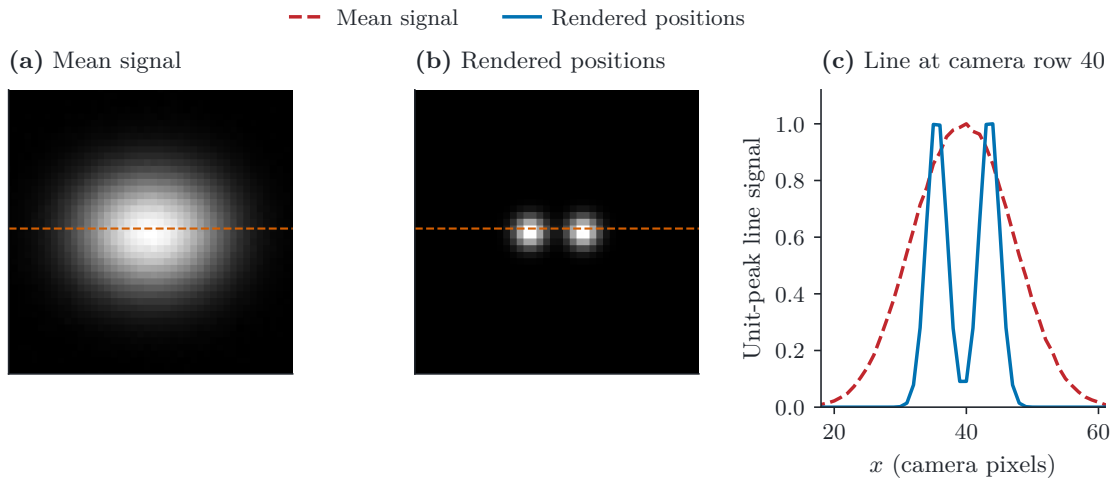
Caption. Current results browser before a reconstruction is run. This is a UI screenshot, not experimental evidence.

Document locations. User guide, Figure 7

Evidence type. actual application screenshot

User guide – Figure 8

Linked image profiles



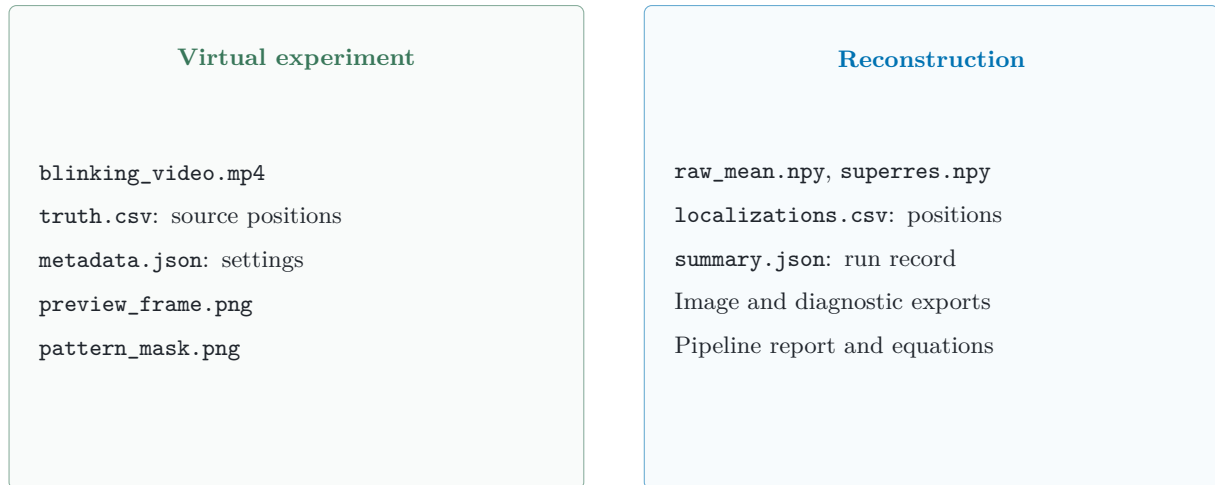
Caption. Linked row profiles from the recorded two-source synthetic reconstruction. Dashed horizontal lines identify camera row 40; each profile is peak-normalized. A profile describes the displayed images and is not an independent resolution measurement.

Document locations. User guide, Figure 8

Evidence type. recorded two-source synthetic application run

User guide – Figure 9

Output files



Caption. Output roles for the virtual experiment and reconstruction workflows.

Document locations. User guide, Figure 9

Evidence type. native TikZ explanatory schematic

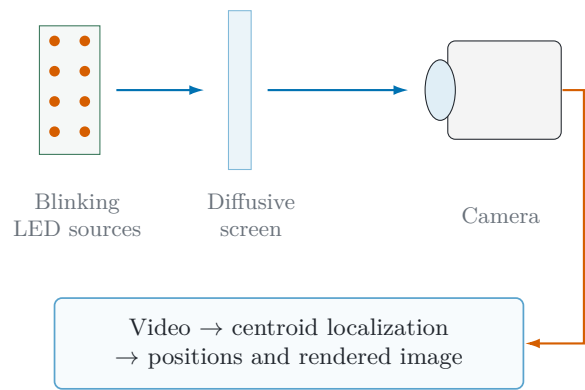
Manuscript – Figure 5

Apparatus and optical pathway

(a) Supplied apparatus design



(b) Optical and data paths



Conceptual diagram; not to scale.
Effective blur and object-plane pixel size require calibration.

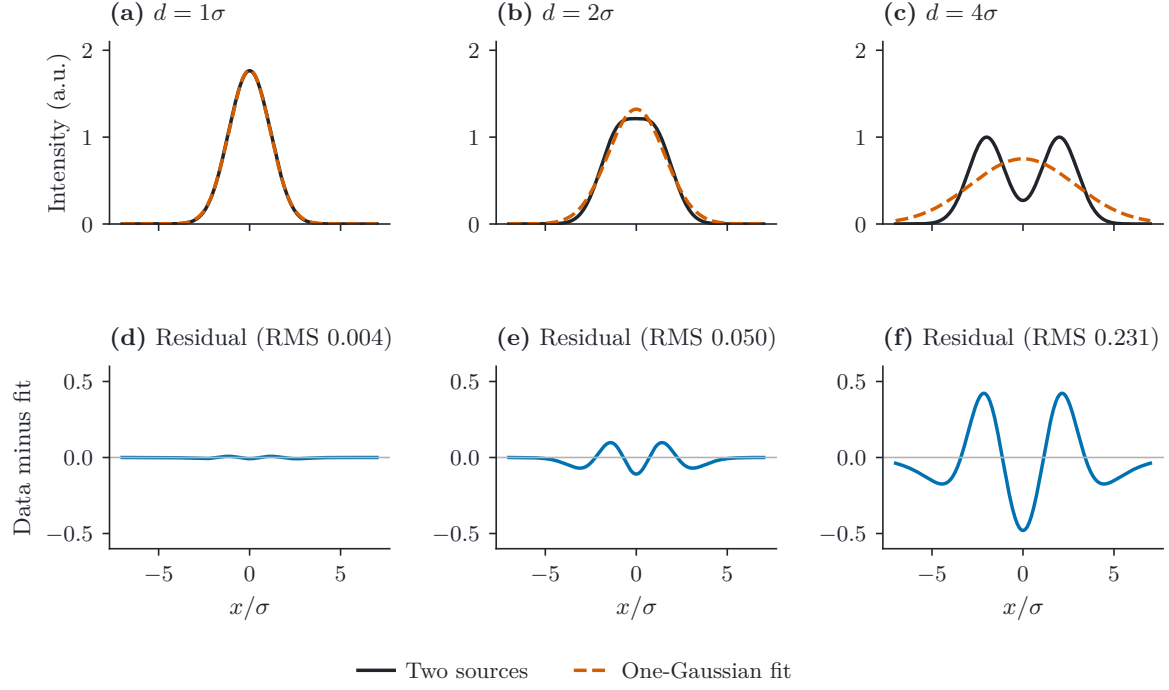
Caption. Supplied apparatus CAD image and a conceptual optical/data pathway. Dimensions and effective blur must be calibrated; no raw-camera inset is claimed.

Document locations. Manuscript, Figure 5

Evidence type. native TikZ explanatory schematic

Manuscript – Figure 6

Single-Gaussian model mismatch



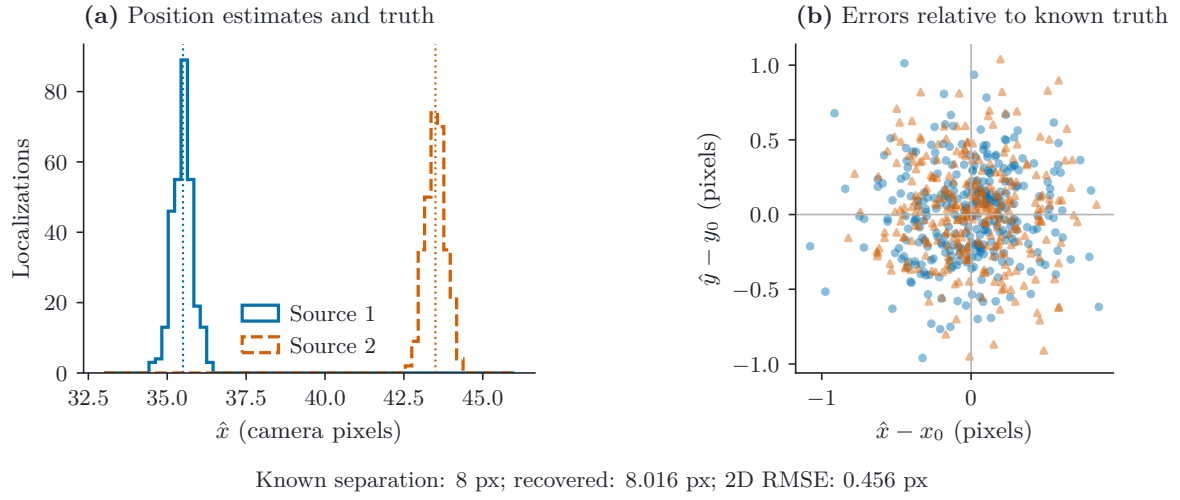
Caption. Analytical two-Gaussian signals fitted by a constrained symmetric single Gaussian. Residuals reveal model mismatch as separation increases. This illustrative fit is not an implemented Classroom-STORM fitting stage, and no automatic acceptance criterion is claimed.

Document locations. Manuscript, Figure 6

Evidence type. analytical or synthetic teaching example

Manuscript – Figure 10

Position validation against truth

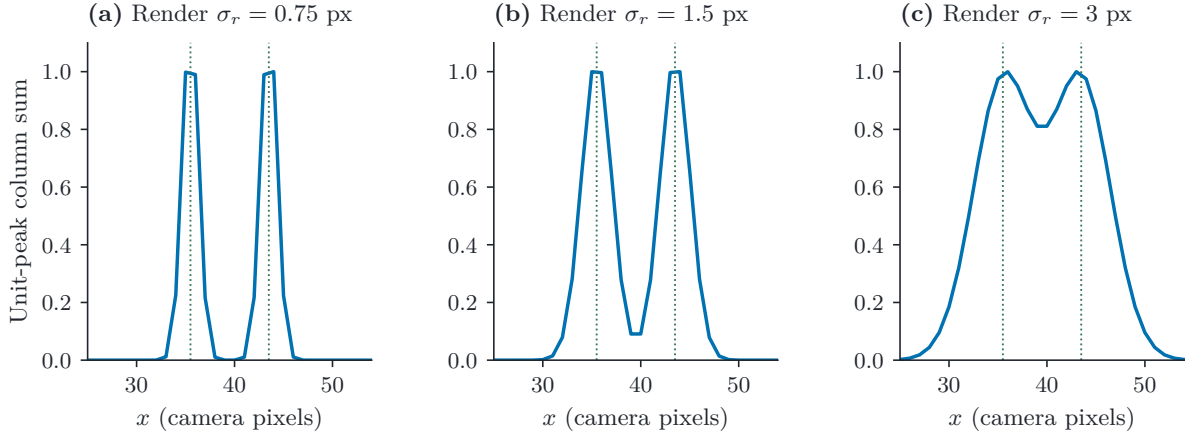


Caption. Position estimates from the recorded 600-frame two-source synthetic run. Source membership comes from recorded simulation truth, not an inferred clustering result. Vertical dotted lines mark true x coordinates. The error scatter and two-dimensional RMSE assess position accuracy independently of the rendering kernel.

Document locations. Manuscript, Figure 10

Evidence type. recorded synthetic application run with known truth

Manuscript – Figure 11
Effect of display-kernel width



Identical estimated positions; only the display kernel changes.

Caption. The same localization list rendered by the application with Gaussian $\sigma_r = 0.75$, 1.5 or 3 camera pixels. Column-sum profiles are independently peak-normalized. Narrower display kernels change apparent peak widths and valley depths without improving the underlying position estimates or establishing structural resolution.

Document locations. Manuscript, Figure 11

Evidence type. controlled rendering variation on one recorded synthetic localization list